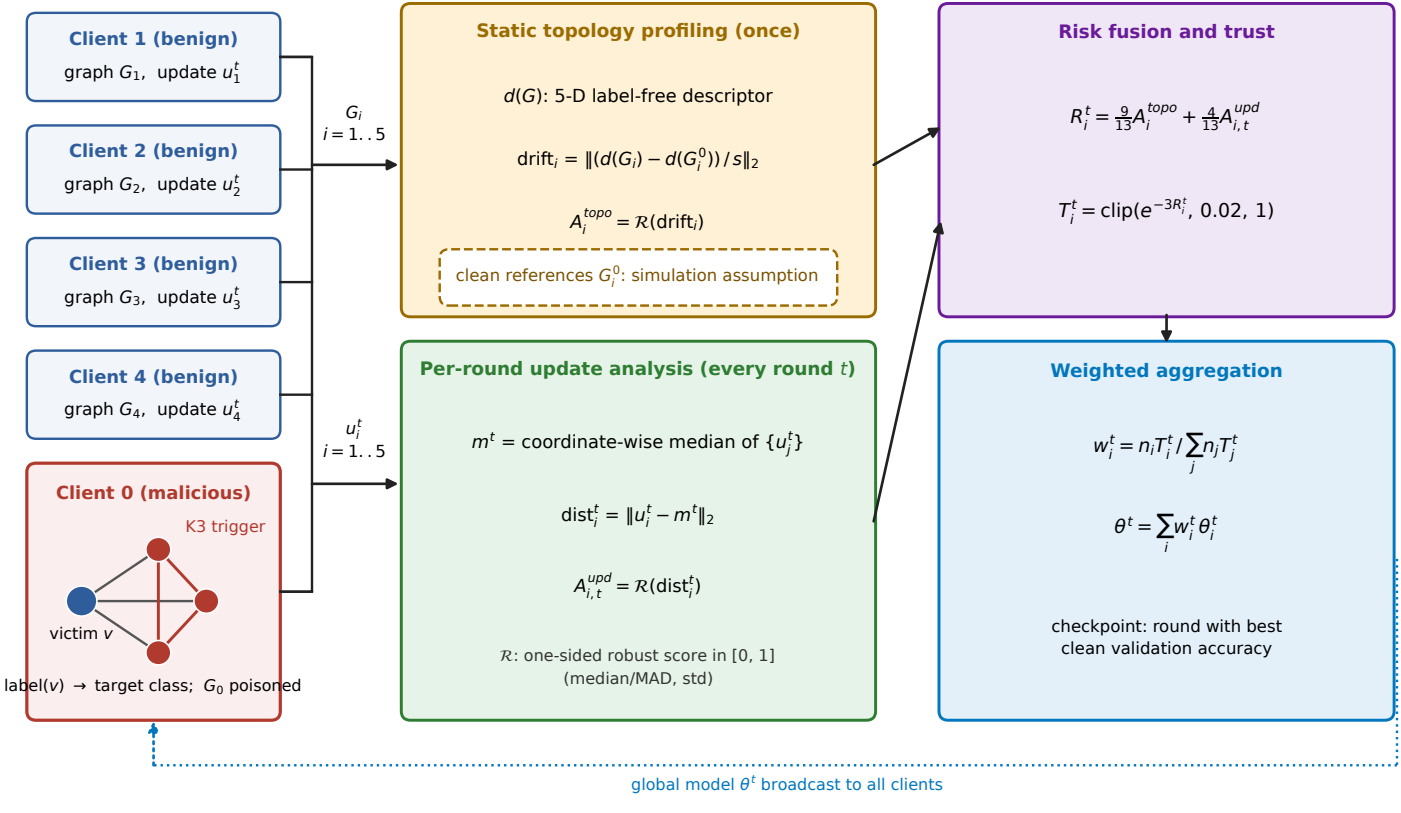
Client 1 (benign)
graph G_1, update u_1^t
Client 2 (benign)
graph G_2, update u_2^t
Client 3 (benign)
graph G_3, update u_3^t
Client 4 (benign)
graph G_4, update u_4^t
Client 0 (malicious)
K3 trigger
victim v
label(v) → target class; G_0 poisoned
G_i
$i = 1..5$
u_i^t
$i = 1..5$
Static topology profiling (once)
$d(G)$: 5-D label-free descriptor
$\text{drift}_i = \|(d(G_i) - d(G_i^0))/s\|_2$
$A_i^{topo} = \mathcal{R}(\text{drift}_i)$
clean references G_i^0: simulation assumption
Per-round update analysis (every round t)
m^t = coordinate-wise median of $\{u_j^t\}$
$\text{dist}_i^t = \|u_i^t - m^t\|_2$
$A_{i,t}^{upd} = \mathcal{R}(\text{dist}_i^t)$
$\mathcal{R}$: one-sided robust score in [0, 1]
(median/MAD, std)
Risk fusion and trust
$R_i^t = \frac{9}{13}A_i^{topo} + \frac{4}{13}A_{i,t}^{upd}$
$T_i^t = \text{clip}(e^{-3R_i^t}, 0.02, 1)$
Weighted aggregation
$w_i^t = n_i T_i^t / \sum_j n_j T_j^t$
$\theta^t = \sum_i w_i^t \theta_i^t$
checkpoint: round with best
clean validation accuracy
global model θ^t broadcast to all clients